CELT Reference Manual

0.3.2

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Chapter 1

CELT Module Index

1.1 CELT Modules

Here is a list of all modules:

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Chapter 2

CELT Data Structure Index

2.1 CELT Data Structures

Here are the data structures with brief descriptions:

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Chapter 3

CELT File Index

3.1 CELT File List

Here is a list of all documented files with brief descriptions:

libcelt/celt.h (Contains all the functions for encoding and decoding audio streams) 15
libcelt/**celt_header.h** . . . . . **??**
libcelt/celt_types.h (CELT types) 18

Chapter 4

CELT Module Documentation

4.1 Encoding and decoding

Functions

- CELTMode * celt_mode_create (celt_int32_t Fs, int channels, int frame_size, int lookahead, int *error)
- void celt_mode_destroy (CELTMode *mode)
- int celt_mode_info (const CELTMode *mode, int request, celt_int32_t *value)
- CELTEncoder * celt_encoder_create (const CELTMode *mode)
- void celt_encoder_destroy (CELTEncoder *st)
- int celt_encode (CELTEncoder *st, celt_int16_t *pcm, unsigned char *compressed, int nbCompressedBytes)
- CELTDecoder * celt_decoder_create (const CELTMode *mode)
- void celt_decoder_destroy (CELTDecoder *st)
- int celt_decode (CELTDecoder *st, unsigned char *data, int len, celt_int16_t *pcm)

4.1.1 Detailed Description

4.1.2 Function Documentation

4.1.2.1 int celt_decode (CELTDecoder * *st*, unsigned char * *data*, int *len*, celt_int16_t * *pcm*)

Decodes a frame of audio.

Parameters:

st Decoder state

data Compressed data produced by an encoder

len Number of bytes to read from "data". This MUST be exactly the number of bytes returned by the encoder. Using a larger value WILL NOT WORK.

pcm One frame (frame_size samples per channel) of decoded PCM will be returned here.

Returns:

Error code.

4.1.2.2 CELTDecoder* celt_decoder_create (const CELTMode * *mode*)

Creates a new decoder state. Each stream needs its own decoder state (can't be shared across simultaneous streams).

Parameters:

mode Contains all the information about the characteristics of the stream (must be the same characteristics as used for the encoder)

Returns:

Newly created decoder state.

4.1.2.3 void celt_decoder_destroy (CELTDecoder * *st*)

Destroys a a decoder state.

Parameters:

st Decoder state to be destroyed

4.1.2.4 int celt_encode (CELTEncoder * *st*, celt_int16_t * *pcm*, unsigned char * *compressed*, int *nbCompressedBytes*)

Encodes a frame of audio.

Parameters:

st Encoder state

pcm PCM audio in signed 16-bit format (native endian). There must be exactly frame_size samples per channel. The input data is overwritten by a copy of what the remote decoder would decode.

compressed The compressed data is written here

nbCompressedBytes Number of bytes to use for compressing the frame (can change from one frame to another)

Returns:

Number of bytes written to "compressed". Should be the same as "nbCompressedBytes" unless the stream is VBR. If negative, an error has occured (see error codes). It is IMPORTANT that the length returned be somehow transmitted to the decoder. Otherwise, no decoding is possible.

4.1.2.5 CELTEncoder* celt_encoder_create (const CELTMode * *mode*)

Creates a new encoder state. Each stream needs its own encoder state (can't be shared across simultaneous streams).

Parameters:

mode Contains all the information about the characteristics of the stream (must be the same characteristics as used for the decoder)

Returns:

Newly created encoder state.

4.1.2.6 void celt_encoder_destroy (CELTEncoder * *st*)

Destroys a an encoder state.

Parameters:

st Encoder state to be destroyed

4.1.2.7 CELTMode* celt_mode_create (celt_int32_t *Fs*, int *channels*, int *frame_size*, int *lookahead*, int * *error*)

Creates a new mode struct. This will be passed to an encoder or decoder. The mode MUST NOT BE DESTROYED until the encoders and decoders that use it are destroyed as well.

Parameters:

Fs Sampling rate (32000 to 64000 Hz)

channels Number of channels

frame_size Number of samples (per channel) to encode in each packet (64 - 256)

lookahead Extra latency (in samples per channel) in addition to the frame size (between 32 and frame_size). The larger that value, the better the quality (at the expense of latency)

error Returned error code (if NULL, no error will be returned)

Returns:

A newly created mode

4.1.2.8 void celt_mode_destroy (CELTMode * *mode*)

Destroys a mode struct. Only call this after all encoders and decoders using this mode are destroyed as well.

Parameters:

mode Mode to be destroyed

4.1.2.9 int celt_mode_info (const CELTMode * *mode*, int *request*, celt_int32_t * *value*)

Query information from a mode

Chapter 5

CELT Data Structure Documentation

5.1 CELTHeader Struct Reference

Header data.

`#include <celt_header.h>`

Data Fields

- char codec_id [8]
- char codec_version [20]
- celt_int32_t version_id
- celt_int32_t header_size
- celt_int32_t sample_rate
- celt_int32_t nb_channels
- celt_int32_t frame_size
- celt_int32_t overlap
- celt_int32_t bytes_per_packet
- celt_int32_t extra_headers

5.1.1 Detailed Description

Header data.

Header data to be used for Ogg files (or possibly other encapsulation)

Definition at line 45 of file celt_header.h.

5.1.2 Field Documentation

5.1.2.1 char CELTHeader::codec_id[8]

MUST be "CELT " (four spaces)

Definition at line 46 of file celt_header.h.

5.1.2.2 char CELTHeader::codec_version[20]

Version used (as string)

Definition at line 47 of file celt_header.h.

5.1.2.3 celt_int32_t CELTHeader::version_id

Version id (negative for until stream is frozen)

Definition at line 48 of file celt_header.h.

5.1.2.4 celt_int32_t CELTHeader::header_size

Size of this header

Definition at line 49 of file celt_header.h.

5.1.2.5 celt_int32_t CELTHeader::sample_rate

Sampling rate of the original audio

Definition at line 50 of file celt_header.h.

5.1.2.6 celt_int32_t CELTHeader::nb_channels

Number of channels

Definition at line 51 of file celt_header.h.

5.1.2.7 celt_int32_t CELTHeader::frame_size

Samples per frame (per channel)

Definition at line 52 of file celt_header.h.

5.1.2.8 celt_int32_t CELTHeader::overlap

Overlapping samples (per channel)

Definition at line 53 of file celt_header.h.

5.1.2.9 celt_int32_t CELTHeader::bytes_per_packet

Number of bytes per compressed packet (0 if unknown)

Definition at line 54 of file celt_header.h.

5.1.2.10 celt_int32_t CELTHeader::extra_headers

Number of additional headers that follow this header

Definition at line 55 of file celt_header.h.

The documentation for this struct was generated from the following file:

- libcelt/celt_header.h

Chapter 6

CELT File Documentation

6.1 libcelt/celt.h File Reference

Contains all the functions for encoding and decoding audio streams.

```
#include "celt_types.h"
```

Defines

- #define CELT_OK 0
- #define CELT_BAD_ARG -1
- #define CELT_INVALID_MODE -2
- #define CELT_INTERNAL_ERROR -3
- #define CELT_CORRUPTED_DATA -4
- #define CELT_GET_FRAME_SIZE 1000
- #define CELT_GET_LOOKAHEAD 1001
- #define CELT_GET_NB_CHANNELS 1002
- #define CELT_GET_BITSTREAM_VERSION 2000

Typedefs

- typedef struct CELTEncoder CELTEncoder

  Encoder state.

- typedef struct CELTDecoder CELTDecoder
- typedef struct CELTMode CELTMode

Functions

- CELTMode * celt_mode_create (celt_int32_t Fs, int channels, int frame_size, int lookahead, int *error)
- void celt_mode_destroy (CELTMode *mode)
- int celt_mode_info (const CELTMode *mode, int request, celt_int32_t *value)
- CELTEncoder * celt_encoder_create (const CELTMode *mode)
- void celt_encoder_destroy (CELTEncoder *st)

- int celt_encode (CELTEncoder *st, celt_int16_t *pcm, unsigned char *compressed, int nbCompressedBytes)
- CELTDecoder * celt_decoder_create (const CELTMode *mode)
- void celt_decoder_destroy (CELTDecoder *st)
- int celt_decode (CELTDecoder *st, unsigned char *data, int len, celt_int16_t *pcm)

6.1.1 Detailed Description

Contains all the functions for encoding and decoding audio streams.

Definition in file celt.h.

6.1.2 Define Documentation

6.1.2.1 #define CELT_BAD_ARG -1

An (or more) invalid argument (e.g. out of range)

Definition at line 50 of file celt.h.

6.1.2.2 #define CELT_CORRUPTED_DATA -4

The data passed (e.g. compressed data to decoder) is corrupted

Definition at line 56 of file celt.h.

6.1.2.3 #define CELT_GET_BITSTREAM_VERSION 2000

GET the bit-stream version for compatibility check

Definition at line 67 of file celt.h.

6.1.2.4 #define CELT_GET_FRAME_SIZE 1000

GET the frame size used in the current mode

Definition at line 60 of file celt.h.

6.1.2.5 #define CELT_GET_LOOKAHEAD 1001

GET the lookahead used in the current mode

Definition at line 62 of file celt.h.

6.1.2.6 #define CELT_GET_NB_CHANNELS 1002

GET the number of channels used in the current mode

Definition at line 64 of file celt.h.

6.1.2.7 #define CELT_INTERNAL_ERROR -3

An internal error was detected

Definition at line 54 of file celt.h.

6.1.2.8 #define CELT_INVALID_MODE -2

The mode struct passed is invalid

Definition at line 52 of file celt.h.

6.1.2.9 #define CELT_OK 0

No error

Definition at line 48 of file celt.h.

6.1.3 Typedef Documentation

6.1.3.1 typedef struct CELTDecoder CELTDecoder

State of the decoder. One decoder state is needed for each stream. It is initialised once at the beginning of the stream. Do ∗not∗ re-initialise the state for every frame

Definition at line 80 of file celt.h.

6.1.3.2 typedef struct CELTEncoder CELTEncoder

Encoder state.

Contains the state of an encoder. One encoder state is needed for each stream. It is initialised once at the beginning of the stream. Do ∗not∗ re-initialise the state for every frame.

Definition at line 75 of file celt.h.

6.1.3.3 typedef struct CELTMode CELTMode

The mode contains all the information necessary to create an encoder. Both the encoder and decoder need to be initialised with exactly the same mode, otherwise the quality will be very bad

Definition at line 85 of file celt.h.

6.2 libcelt/celt_types.h File Reference

CELT types.

Typedefs

- typedef short celt_int16_t
- typedef unsigned short celt_uint16_t
- typedef int celt_int32_t
- typedef unsigned int celt_uint32_t
- typedef long long celt_int64_t
- typedef unsigned long long celt_uint64_t

6.2.1 Detailed Description

CELT types.

Definition in file celt_types.h.

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